

Software Test Document

Project: Home Network Solutions

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1. Introduction

1.1 Purpose

This document describes how the Home Network solutions project will be tested and the success criteria it will need to pass. The features that will be tested will have certain specific and measurable criteria that they must pass in order to be considered complete. Not all features will be tested, some due to the nature of the feature, priority, or complexity of the feature. Additionally, performance and security will be assessed, as both are a priority

1.2 Scope

Testing focuses on functionality and usability, and will cover the server setup process, the web application, user authentication, setup wizard, dashboard, file management, integration of optional applications, and SQLite database.

1.3 References

- Tailscale Documentation: <https://tailscale.com/docs>
- Jellyfin Documentation: <https://jellyfin.org/docs/>
- Pi-hole Documentation: <https://docs.pi-hole.net/>
- Home Assistant Documentation: <https://www.home-assistant.io/docs/>
- Caddy Documentation: <https://caddyserver.com/docs/>
- Git Documentation: <https://git-scm.com/docs>
- Samba Documentation: <https://www.samba.org/samba/docs/>
- Jira Software Documentation: <https://support.atlassian.com/jira-software-cloud/resources/>

2. Test Items

The following items will be tested:

- Server setup script
- Web application
- File upload/download system
- SQLite database

3. Features to be Tested

The following features will be tested:

- Login authentication
- Setup wizard
- Dashboard behavior
- File upload/download accessibility
- Integration of Pi-hole, Tailscale, Home Assistant, and Jellyfin
- Database operations

4. Test Approach

The use of unit testing, integration testing, system testing, and usability testing will ensure that all server and web app features function properly and provide a seamless user experience for both technical and non-technical users.

4.1 Unit Testing

The following functions within the web app will be tested in isolation, given both pass and fail inputs:

- Password hashing
- Session token generation
- Database read/write

These tests will be implemented using Python's pytest framework with each unit test calling a single function directly with varied inputs. Tests will be organized into a tests/ directory and run locally before each main branch merge.

4.2 Integration Testing

Integration testing will ensure that the SQLite database, file system, and services such as Pi-hole, Tailscale, Home Assistant, and Jellyfin correctly link to the web app. This will be done by:

- Sending HTTP requests to the web app's endpoints to verify the database and filesystem state
- Calling each service's API to confirm configuration

These tests will be run on a staging instance of the server to avoid interfering with the actual deployment.

4.3 System Testing

The system will be tested on a Raspberry Pi 5 with four gigabytes of ram, covering the initial setup from a blank install, logging in and accessing the dashboard, and uploading a file remotely before confirming it is accessible locally. This will be done manually to ensure tasks such as the initial hardware setup wizard are completed correctly, as well as ensuring the device can fully communicate with the server and vice versa.

4.4 Usability Testing

A group of testers unfamiliar with the project will be asked to complete core tasks such as initial setup, adding a widget, and uploading a file without assistance. Each tester will have their screen recorded to see how long each task takes and if there were any hesitations, backtracks, or questions asked. Each tester will be given a questionnaire rating the difficulty of each task.

This combination of questions and screen recordings will give us good insight into the users' experiences. The screen recordings will help demonstrate what they instinctively think about certain elements, such as which features should be on which pages and where buttons and links should be located, whereas the questionnaire will allow them to give more direct feedback on things the screen recording might not catch.